

MI&CT eSERVICE NOTE

SN08011412a

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This short-term informational note is invalid 6 months after the issue date.

Date: Jan. 14, 2008

Subject: Possible Wiring Error Between Slip Ring and OBC Chassis

Effectivity: All systems utilizing the H2, H3 and H16 gantries (LightSpeed 2X-4X)

NOTE: FMI 25399 will inspect a select group of the CT LightSpeed systems manufactured in Hino, Japan between June 1st 2002 and December 31st 2002, which have H2 and H3 CT gantries (LightSpeed 2X-3X). These systems are more likely to have the wiring error.

Details:

There is a possibility that CT LightSpeed 2X – 4X gantries may contain a wiring error, which reverses the 120 volt hot and neutral lines between the Slip Ring and the OBC chassis. The mis-labeling of the two wires by the Supplier caused this mis-wire in the gantry manufacturing process. The mis-wire error alone does not cause direct faults with the system, but indirectly can cause damage to gantry electrical components (mainly OBC Power I/F Board) if a short circuit occurs on the 120 volt hot line. With the 120 volt hot and neutral lines reversed, the hot line is no longer fuse protected, allowing 40 amps to flow through the OBC Power I/F Board traces.

Resolution:

On the next PM visit or when service is required on the gantry, use the following procedure to inspect for this wiring error. This Service Note will explain the procedure one should follow to inspect the gantry.

Procedure

Removing Gantry and Slip Ring Covers

1. Remove the Gantry side covers.



NOTICE

Always turn OFF the HVDC before the 120 VAC. Turning OFF 120 VAC power before HVDC power can result in equipment damage.

2. Turn OFF Axial Enable, HVDC, and 120VAC Service switches to disable Gantry motion.
3. Position OBC Chassis at 2 o'clock.

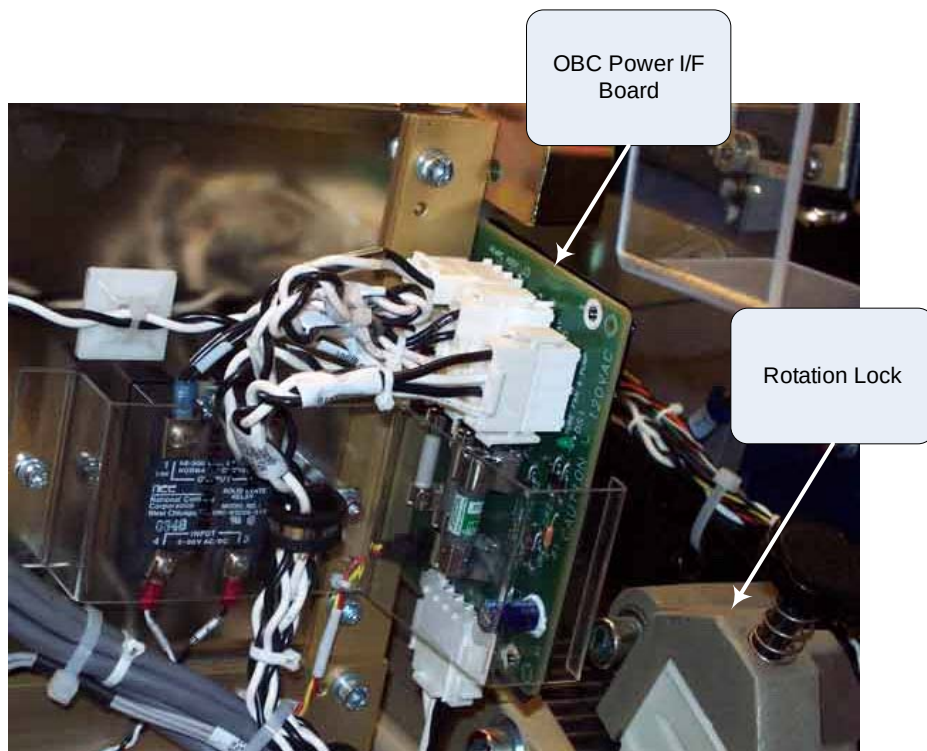


Figure 1: OBC Power I/F Board Location

With OBC Chassis at 2 O'clock position the OBC Power I/F Board will be located close to the Rotational Lock Assembly.

⚠ DANGER
ELECTROCUTION HAZARD.
HIGH VOLTAGE PRESENT.
USE PROPER LOCKOUT/TAGOUT PROCEDURES BEFORE REPLACING COMPONENTS.

4. Completely shutdown power at A1 panel, lock out and tag equipment.
5. Remove the Gantry Top and Back Covers to gain access to the Slip Ring Assembly. Follow cover removal procedures in Service Manual as needed.

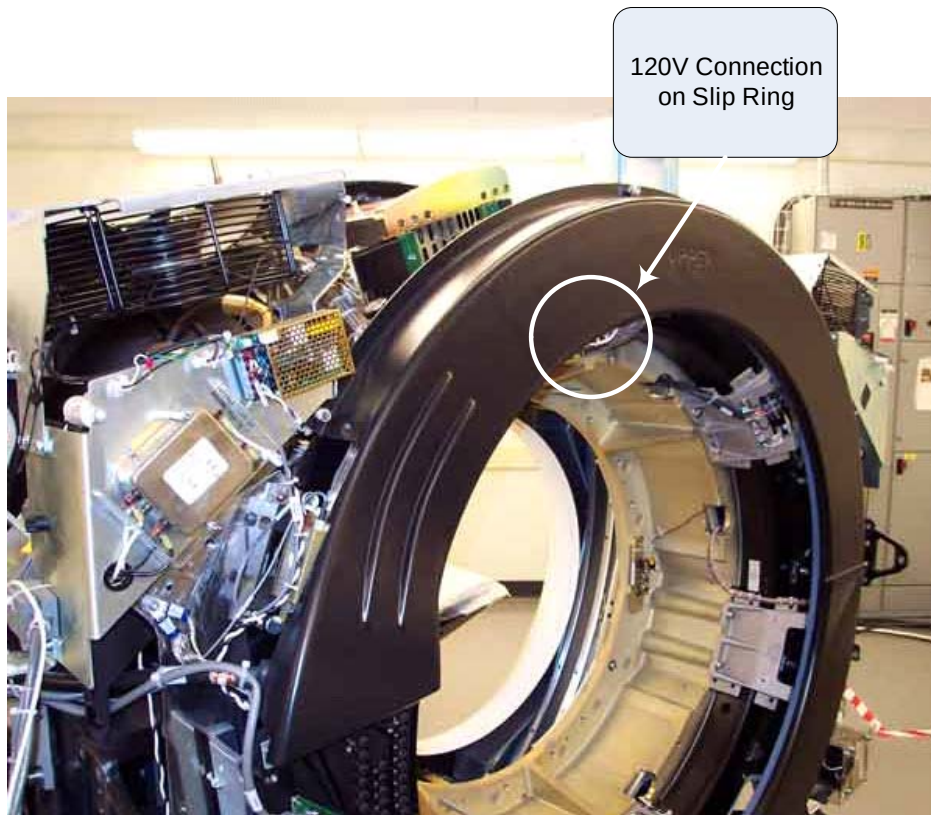


Figure 2: 120V Connection on Slip Ring Location

With OBC Chassis at 2 O'clock position the 120V Slip Ring connection will be located at the top left side of gantry viewed from the rear.

6. Remove the Upper Slip Ring Cover and set aside.

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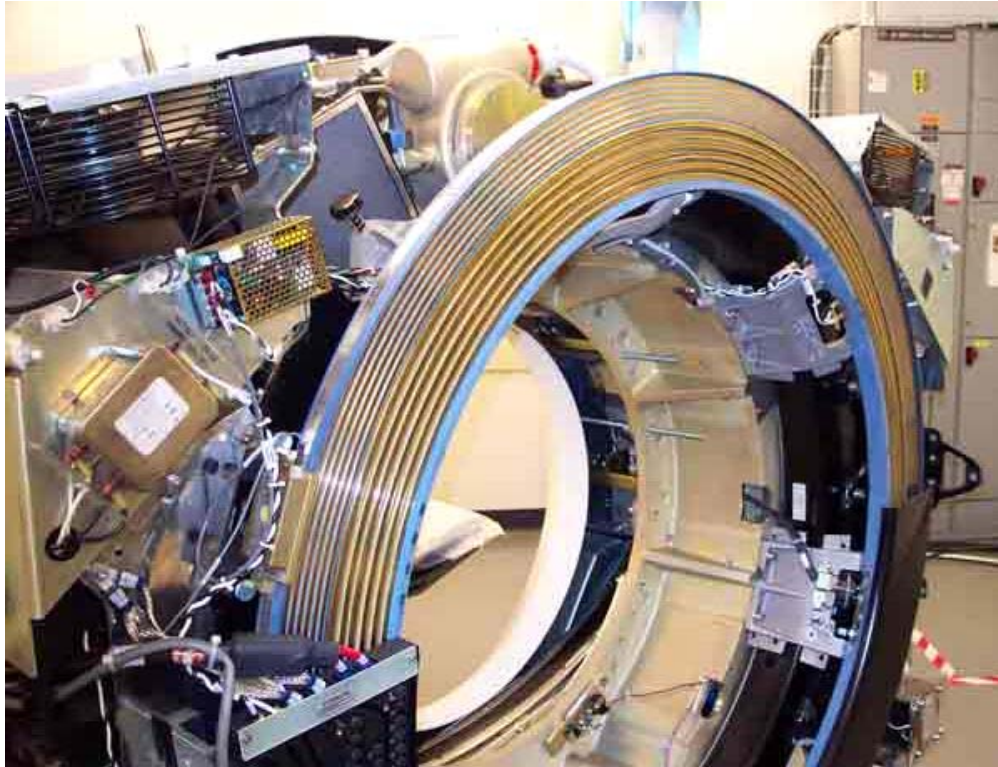


Figure 3: Upper Cover removed from Slip Ring

Inspect 120V Harness (2262943) Connections at Slip Ring

7. Inspect the 120Vac wiring connection on the Slip Ring and compare it to the Figure 4 below.

Note: It is critical to inspect both the Labeling on the Wires and the Connections!

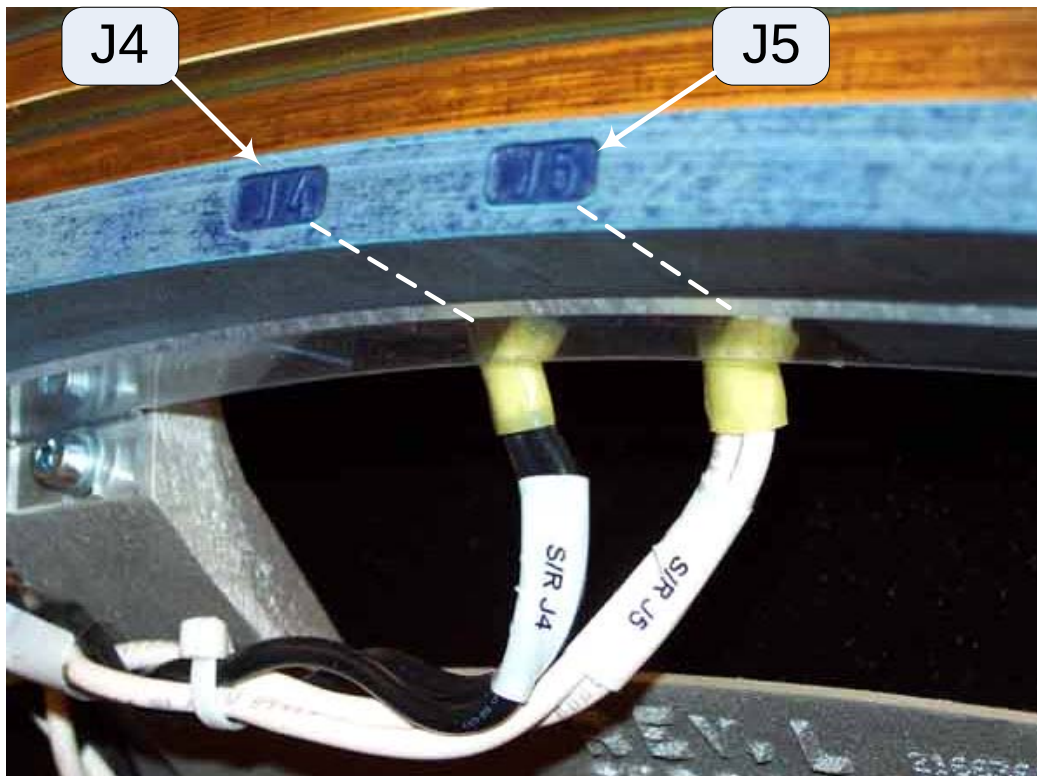


Figure 4: Proper wiring connection on Slip Ring

8. Verify the following wire labeling and connections:
 - a. Black Wire is labeled S/R J4 and is connected to the J4 connection on Slip Ring.
 - b. White Wire is labeled, S/R J5 and is connected to the J5 connection on Slip Ring.
9. If wires are not labeled correctly or are not connected as illustrated above:
 - a. Correct the wiring error.
and/or
 - b. Correct the labels using a permanent marker.
10. Install Slip Ring Cover and Back Gantry Cover.

Electical Check of 120Vac Harness

11. Remove Lock /Tag and turn on system power at A1 panel.
12. Turn ON 120VAC Service switch.
13. Using a Multimeter (DVM) set to measure AC Voltage, place the positive probe on the large fuse located on the OBC Power I/F Board and the negative probe on chassis ground.

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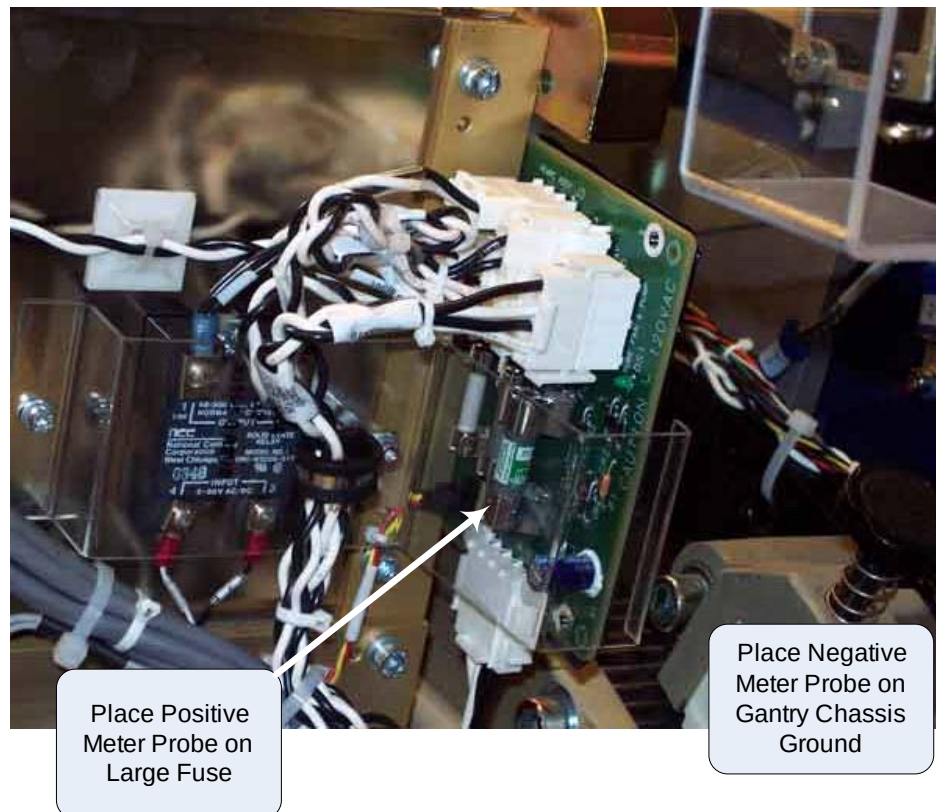


Figure 5: Electrical Check for Proper Wiring

14. Verify that DVM reads 120VAC. If it does not measure 120VAC, ensure 120VAC is enabled or recheck wiring to Slip Ring. Troubleshoot as needed.
15. Install Gantry Top Covers.
16. Turn ON Axial Enable and HVDC Service switches to enable gantry rotation.
17. Install the gantry side covers.
18. Perform a test scan to confirm system performance before returning system to the customer.

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Affected Service Publication:

- None

PQR/SPR #:

- PSR 13085844 & 13099568

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